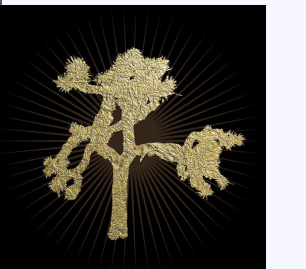
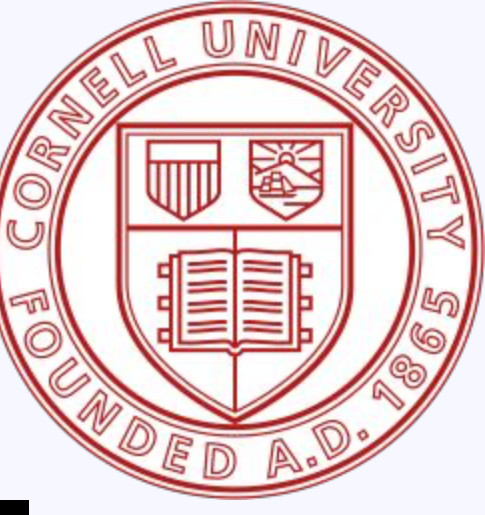


Compressive MRI Reconstruction via FISTA and Plug-and-Play Priors

Griffin Carter, Wallace Wei



Introduction

We use optimization (FISTA) and Machine Learning (PnP Priors) to improve our Compressive MRI Reconstruction in both convergence and accuracy

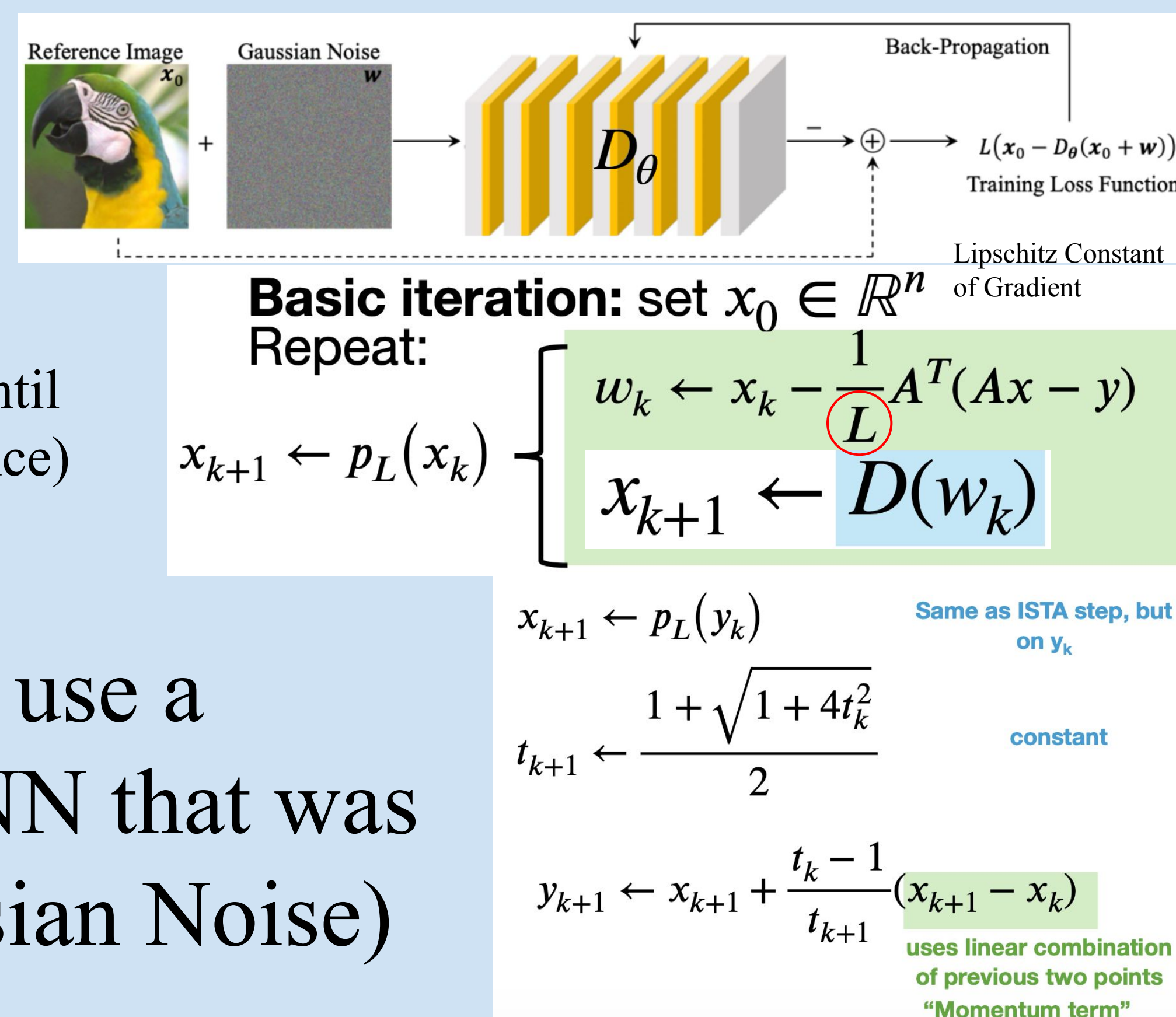
Methods

We used **PnP-FISTA**, which repeats three basic steps

1. Data Fidelity Step
2. Denoiser Step
3. Momentum Step

(Repeat Until Convergence)

Denoiser (PnP Prior) → We use a **DRU-Net** (Prebuilt U-Net CNN that was trained on Images with Gaussian Noise)



Results

To aid with convergence, we modified our **PnP Denoiser** to become a **contractive denoiser** (guarantees fixed point convergence)

$$\|D(x) - D(y)\| \leq L\|x - y\|, \quad L < 1 \quad D_n(x_k) = x_k + L(\beta(Pnp(x_k) - x_k))$$

We conducted a parameter sweep where they were chosen by minimizing relative errors (using the correct sigma is very important for FISTA + PnP)

(Normal) FISTA uses a <u>convex TV prior</u>	Convergence (iterations)	Relative Error $rel_error = \frac{\ \hat{x} - x\ _2}{\ x\ _2}$
POCS	20	0.096
FISTA	~300	0.075
FISTA + PnP	~30	0.067

FISTA + PnP uses less iterations (than FISTA), but may take longer to converge (simple math v. CNN denoiser). But, this prior is more informative, so it converges to a better solution.

Conclusion

Our methods, while converging slower showed significantly less error, which would improve the accuracy of CS MRI scans.

Further research could look into **Unrolled Networks** (treating FISTA iterations as a Neural Network) or using a **better denoiser** (trained on data like MRI, or experimentation with parameters)

Technical Background

Magnetic Resonance Imaging (MRI) uses magnetic fields & RF to sample k-space (Fourier Domain) which are then transformed into images (like organs, tissues, bones). Compressive Sensing (CS) speeds up MRI by undersampling k-space and exploiting sparsity, (our baseline reconstruction is POCS)

Forward Model → $y = A\Psi s + \epsilon$ $x = \Psi s$

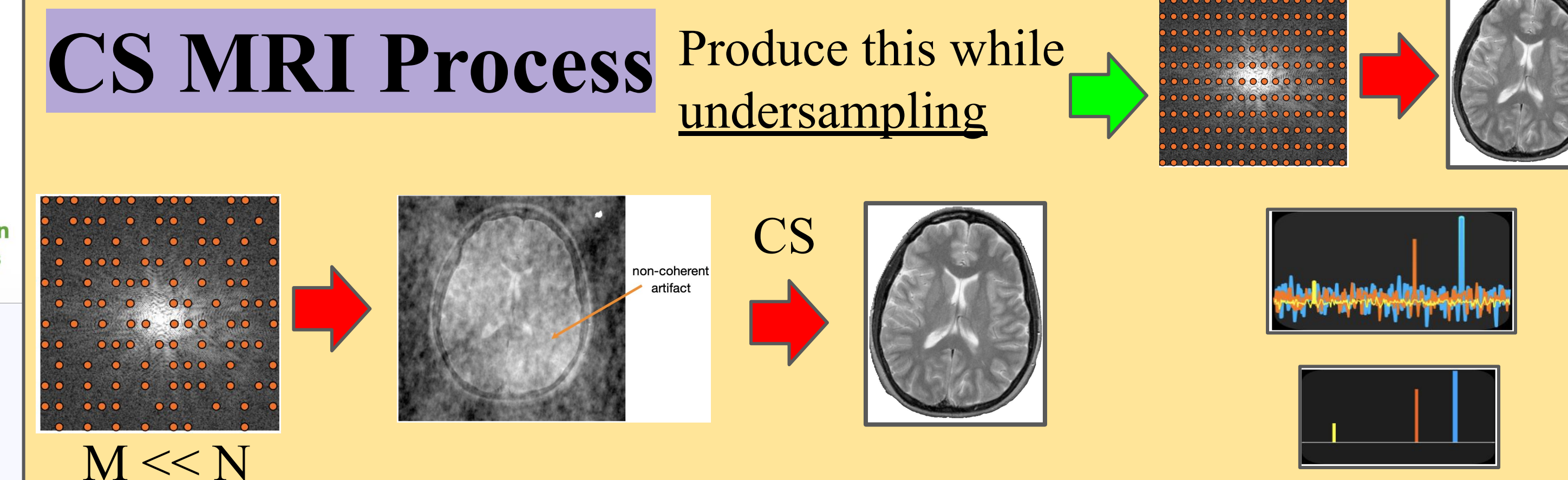
Inverse Problem → $\min_s \frac{1}{2} \|A\Psi s - y\|_2^2 + \lambda R(s)$

POCS (Projection over Convex Sets) $A = M^T$

1. Soft Thresholding Step
 2. Data Fidelity Step
- (Repeat Until Convergence)

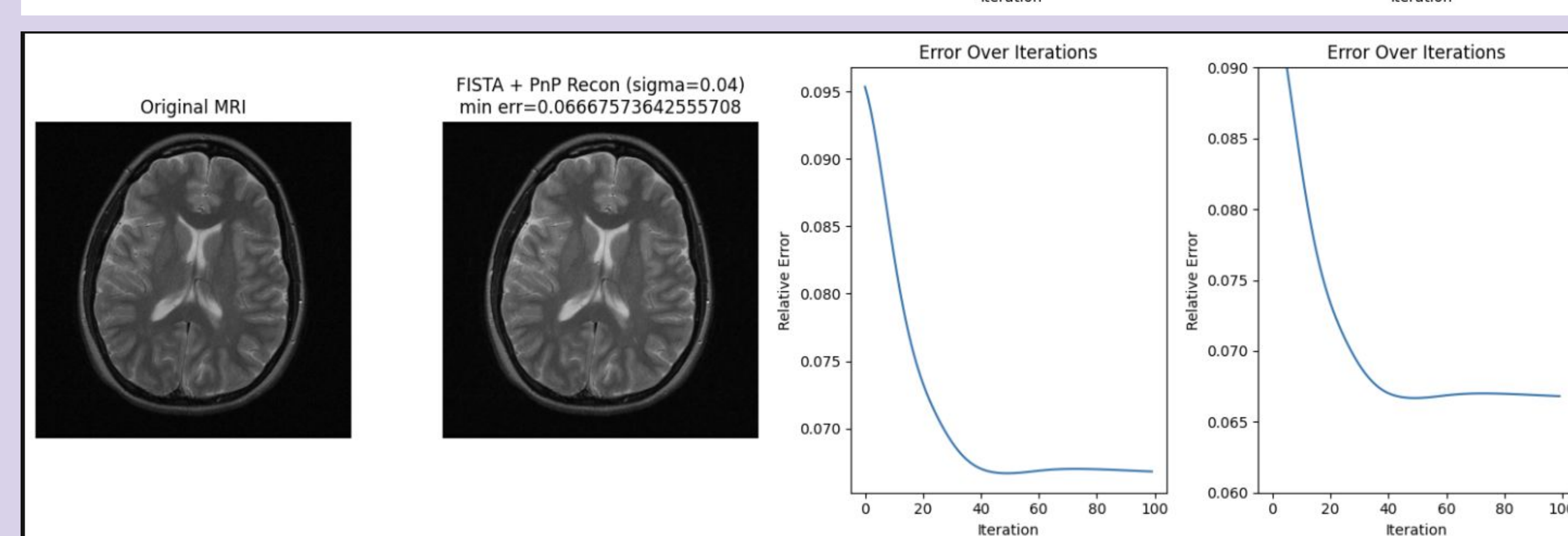
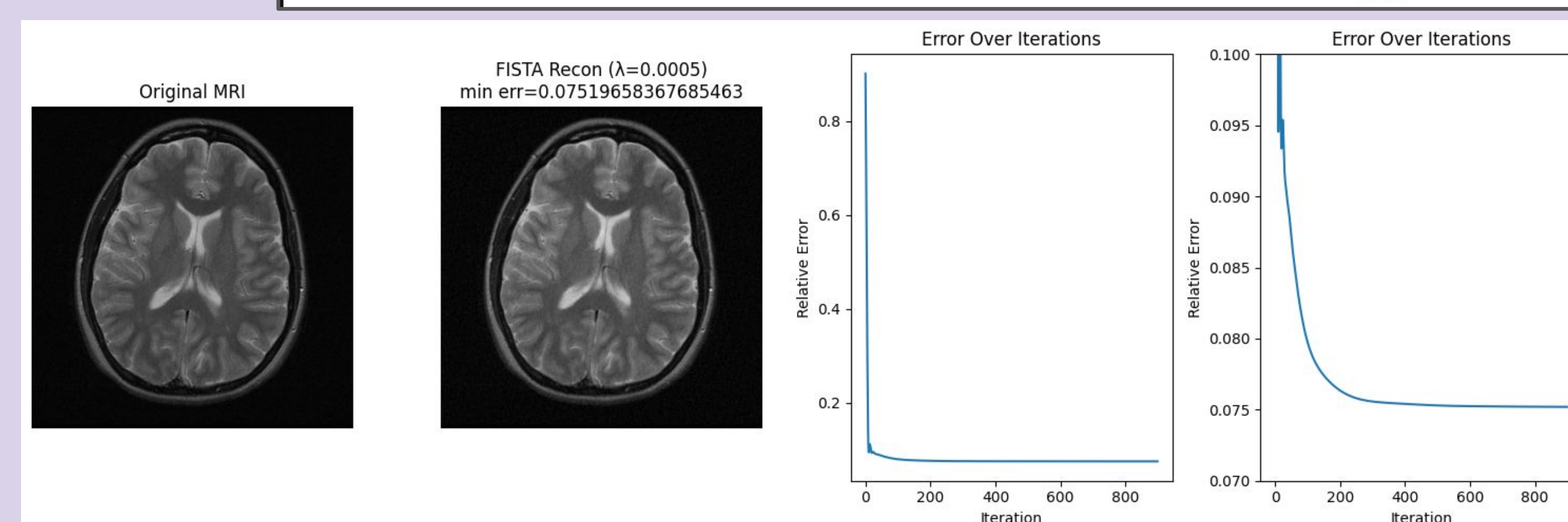
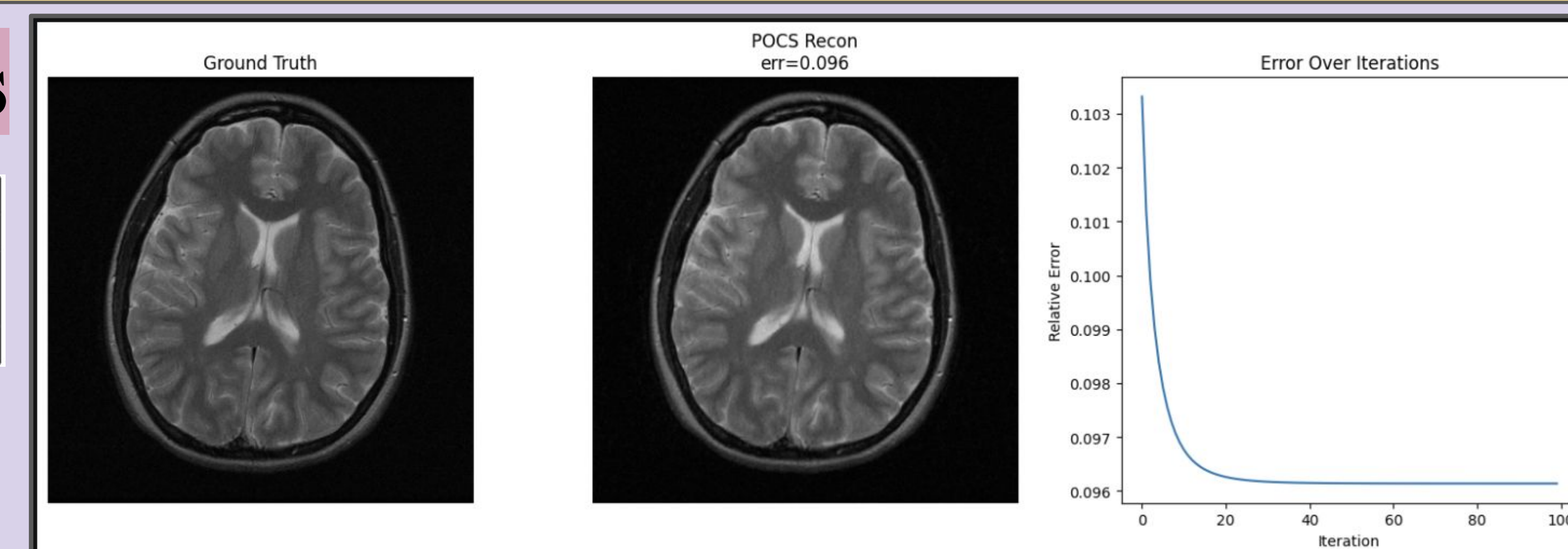
- Let $\tilde{X} = F\tilde{x}$. Initially set $\tilde{X}_0 = Y$.
1. Compute inverse FT to get signal estimate $\tilde{x}_i = F^* \tilde{X}_i$
 2. Apply SoftThresh $\tilde{x}_i = S(\tilde{x}_i, \lambda)$ in the signal domain
 3. Compute the FT $\tilde{X}_i = F\tilde{x}_i$
 4. Enforce data consistency in the frequency domain

$$\tilde{X}_{i+1}[j] = \begin{cases} \tilde{X}_i[j] & \text{if } Y[j] = 0 \\ Y[j] & \text{otherwise} \end{cases}$$



Parameters

Symbol	POCS	FISTA	FISTA + PnP
L	N/A	N/A	0.9
λ	0.03	.0005	0.04
γ	N/A	N/A	0.2
β	N/A	N/A	0.3



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Compressive MRI Reconstruction via FISTA and Plug-and-Play Priors

By Griffin Carter, Wallace Wei

Abstract—Compressive Sensing (CS) in Magnetic Resonance Imaging (MRI) allows for a significant reduction in the time it takes to perform a scan by undersampling the k-space scanned by the MRI. The quality of an CS MRI image depends heavily on the ability to reconstruct with an incomplete dataset. Different implementations of inverse solvers have been used in an attempt to minimize the error from the reconstructed image. Here we present an improved reconstruction algorithm that uses an improved denoising with Plug-and-Play priors, alongside an optimized Fast Iterative Shrinkage-Thresholding Algorithm (FISTA). We found that implementing this FISTA-PnP approach not only achieved faster convergence, but also had a reduction in error compared to traditional wavelet-based POCS methods.

1 INTRODUCTION

MAGNETIC Resonance Imaging (MRI) as a diagnostic imaging technique that uses strong magnets in order to create images of tissues, organs and bones. MRI's use strong magnetic fields in order to excited hydrogen protons in the body. These excited protons will releas signals when as they relax, which can be measured by a scanner to form images based on tissue water content [1]. These images are acquired in the k-space, a spatial frequency domain that captures the raw radiofrequency signals emitted by nuclei when subjected under a magnetic field. This data can be converted into an image by using the Inverse Fourier Transform. However, due to the MRI measuring using the k-space, it is inherently slow to acquire a full sample in the k-space required to reconstruct the image perfectly. Compressive Sensing (CS) aims to reconstruct MRI images from significantly fewer measurements, allowing for a speedup in scan time. This form of MRI reconstruction exploits the fact that the MRI image is very sparse in the wavelet domain. Having a sparse image allows us to store much more information about an image in much fewer pixels. In the case of MRI, this sparsity allows for measurements to under-sample the k-space for our image. This presents a problem, where the under-sampled data means that we cannot directly reconstruct the image from the wavelet domain as we lose some information due to the under-sampling. One way to address this would be to ensure the the k-sampling chosen retains the most information possible. Different techniques, such as following the cartesian plane, or following a spiral, all lead to different reconstructions of the final MRI image [2]. However, the accuracy of reconstruction is largely dependent on the algorithm used for the reconstruction. Therefore, algorithms have been developed in order to solve this optimization problem to minimize the error of the reconstruction.

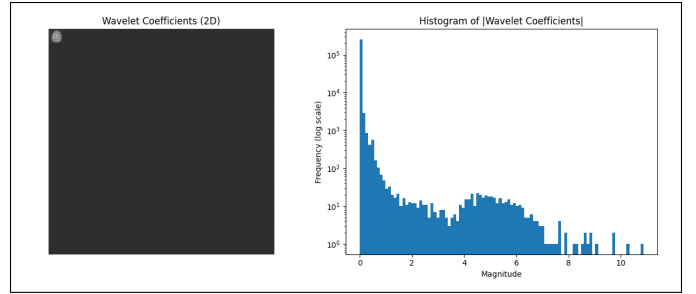


Fig. 1. An MRI image, displayed in the Wavelet Domain

An early attempt to solve this problem utilized Projection onto Convex Sets (POCS). This method iterated between soft-thresholding and constraining data consistency in order to predict the lowest error image given the limited sampling. However, this method has a limited ability to accurately reconstruct the image, where it retains some error even after running for many iterations. In this paper, we explore the optimization of the image solver by introducing a Plug-and-Play-prior for improved denoising of the image alongside the use of FISTA to improve the accuracy of the model.

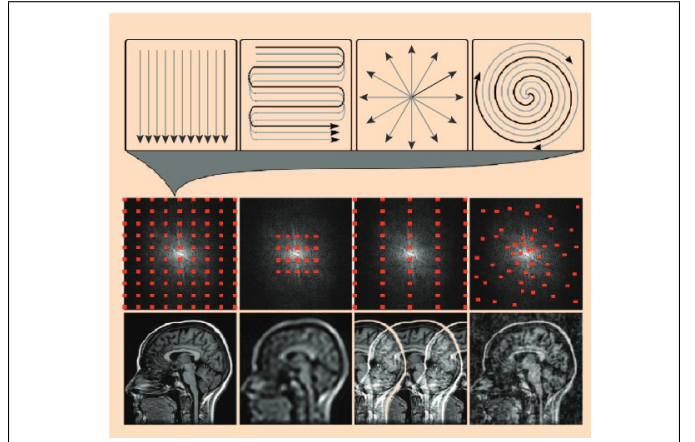


Fig. 2. Various ways to take a subsample of the k-space, and its following affects on the reconstruction

2 FORWARD MODEL

The forward model for CS MRI describes the process of the MRI capturing the k-space image using the scanner. Suppose we have an image $x \in \mathbb{C}^N$ be the original image we wish to recover. We can model the measurement process of an MRI as

$$y = Ax + \eta \quad (1)$$

Where y is the observed undersampled k-space data, A is the forward operator, and η is the White Gaussian Noise. For an MRI, we can define the forward operator as

$$A = M\mathcal{F} \quad (2)$$

Where $\mathcal{F}$ represents the 2D discrete Fourier Transform, and M is a under sampling mask for the k-space. We assume that our image x is sparse in the wavelet domain, such that

$$x = \Psi s \quad (3)$$

Where Ψ is the inverse Wavelet transform and s is some sparse matrix.

3 INVERSE PROBLEM

The inverse problem for the MRI is to find a way to recover image x from measurement y while minimizing the error. In other words, we want the reconstructed image to be a similar as possible to what the true expected image would be. We can define this problem as

$$\hat{x} = \operatorname{argmin}_s D(s, y) + \lambda R(s) \quad (4)$$

Where $R(s)$ is our regularization term (prior), λ is the regularization parameter, and $D(s, y)$ is our cost function, Which we choose to use the regularized least-squares problem

$$D(s, y) = \frac{1}{2} \|\Psi s - y\|_2^2 \quad (5)$$

Due to the under-sampling of the k-space, this problem is underdetermined, which means there are infinite possible solutions. Therefore, the goal is to solve the inverse problem while minimizing the relative error between the predicted image ($\hat{x}$) and the true image (x).

$$\operatorname{rel_error} = \frac{\|\hat{x} - x\|_2}{\|x\|_2} \quad (6)$$

4 METHODS

4.1 POCS Model

The baseline model utilizes the POCS algorithm in order to solve the inverse problem. This model collects two sets constraints that it uses to determine its estimation: Sparsity (through the mathematical prior) and Data Consistency (maintaining the physics of the MRI). The key principle is that in the wavelet domain, we expect the image should be represented sparsely, and that any part of the k-space measured in the MRI should be exactly the same in the scan (i.e any value from the k-space measured should not be

altered in the estimate). The algorithm alternates between these two conditions in order to reduce the error of the image. The following explains how each step is calculated:

- 1) Soft Thresholding (Enforce Sparsity): From the starting image estimate $\hat{X}_0$, in the wavelet domain we ensure that the image is sparse using soft thresholding. The soft thresholding is model as the following:

$$\operatorname{SoftThresh}(x, \lambda) = \operatorname{sign}(x) * \max(|x| - \lambda, 0) \quad (7)$$

This helps with creating sparsity by shrinking all values towards 0. We apply this to the wavelet domain of the MRI scan to enforce sparsity.

$$\hat{X}_i = \mathcal{F}\mathcal{W}^{-1} \operatorname{SoftThresh}(\mathcal{W}(\mathcal{F}^{-1} X_i), \lambda) \quad (8)$$

Where $\mathcal{W}$ represents the transformation into the wavelet domain. This helps denoise the image estimate by ensuring the estimate matches the expected features (namely being sparse) in the wavelet domain.

- 2) Data Fidelity: Since we know the actual values measured by the MRI of some of the k-sample (Y), we ensure that the estimate matches. This is done by replacing any estimate value if it was already measured by the MRI scan, ensuring that the prediction is consistent with the measurement.

$$\hat{X}_{i+1}[j] = \begin{cases} \hat{X}_i[j] & Y[j] = 0 \\ Y[j] & \text{otherwise} \end{cases} \quad (9)$$

(10)

4.2 FISTA Model

This model replaces the POCS algorithm with Fast Iterative Shrinkage-Thresholding Algorithm (FISTA). FISTA is a modification of ISTA that incorporates momentum (which helps to converge faster, with $O(\frac{1}{k^2})$ convergence).

ISTA is a proximal gradient method which alternates between using gradient descent on a data fidelity term (smooth and convex) and a proximal operator (convex but not smooth), which will try to minimize the regularizer while trying to stay close to original data fidelity solution as possible (determined by the tradeoff between), which in this case is a shrinkage thresholding operator $\operatorname{SoftThresh}$, identically used in the previous model.

- 1) Gradient Descent: The standard step for gradient descent is performed. This aims to update parameters to help reduce the cost function by traversing the problem gradient in the opposite direction. Following the gradient down will eventually lead to a minimum in the cost function, thereby finding parameters that minimized the error in the image.

$$w_k = x_k - \frac{1}{L} A^T (Ax - y) \quad (11)$$

$$x_{k+1} = \operatorname{SoftThresh}(w_k, \lambda) \quad (12)$$

- 2) FISTA acceleration: To improve on the convergence rate, FISTA introduces a momentum-based extrapolation sequence in order to “overshoot” in the direction of the solution.

$$t_{k+1} = \frac{1 + \sqrt{1 + 4t_k^2}}{2} \quad (13)$$

$$y_{k+1} = x_{k+1} + \frac{t_k - 1}{t_k}(x_{K+1} - x_k) \quad (14)$$

4.3 FISTA + PnP Model

The proposed model improves on the above model by implements a PnP Priors to help with denoising, alongside the FISTA algorithm to help speed up convergence like in the previous model. PnP Priors are a custom denoiser used to replace the proximal operator inside the FISTA algorithm ($\text{proxg}(v)$ with $f(v)$). This PnP Prior is more “flexible”, as it can directly train on Gaussian Denoising for an arbitrary set of images, and then be used specifically for the denoising step of MRI, using Deep Learning. We used DRUNet for our PnP Priors, which is a U-Net (a specialized Convolutional Neural Network) [3].

The following formulas explain each step this model takes

- 1) Gradient Descent: The same structure for gradient descent in the previous FISTA model is used again
- 2) Plug-and-Play Prior: We apply a pretrained Plug-and-Play denoiser to help reduce error by removing noise and imposing more natural characteristics you would expect in an image. Unlike other methods of image recovery which rely on knowing what a good image looks like (i.e knowing MRI’s are sparse on the wavelet domain), PnP’s train on natural images to learn what natural images should look like. This knowledge gets translated into an iterative process of trying to make a noisy image “more natural”. This prior is passed through a contractive denoiser D_n to help ensure that the entire objective function is closed and convex, with a Lipschitz continuous gradient [4] ($\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$)

$$x_{k+1} = x_{k+1} + \gamma(D_n(x_{k+1}) - x_{k+1}) \quad (15)$$

$$D_n(x_k) = x_k + L(\beta(PnP(x_k, \lambda) - x_k)) \quad (16)$$

- 3) FISTA acceleration: Again, similar to the previous FISTA model, we add an acceleration term to speed up convergence.

5 RESULTS

For each model, we tested through each parameters and took the best model (based on relative error). For POCS and FISTA, since there is only one parameter, we just sweep through it to find the one that produces the minimal relative error.

Since FISTA + PnP has 4 parameters, we could begin by testing around to see what parameters affect it the most (in our experimentation, changing the λ affected it the most.

To rigorously find the best parameters, we used a built in **Bayesian Optimization** (optuna specifically) for about 50 iterations. This is meant to be a black box function optimizer, exactly what we need for a PnP denoiser. We took the best results from the Bayesian Optimization in order to find the best model.

Below are the parameters used in each model

Symbol	POCS	FISTA	FISTA + PnP
L	N/A	N/A	0.9
λ	0.03	.0005	0.04
γ	N/A	N/A	0.2
β	N/A	N/A	0.3

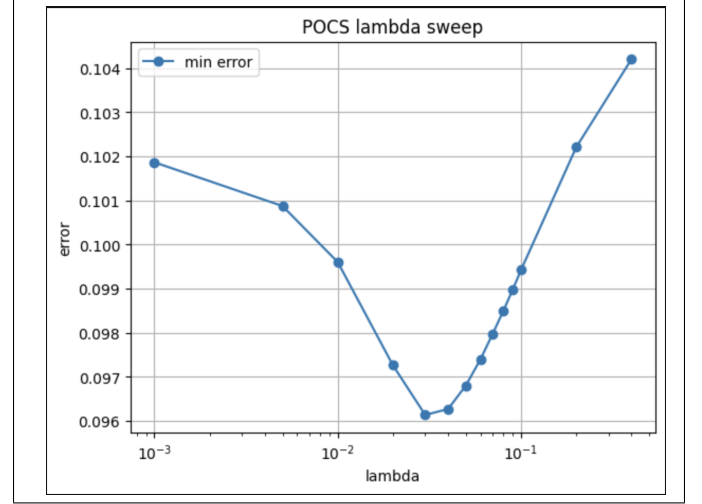


Fig. 3. Relative error of the POCS model for different Lambda values

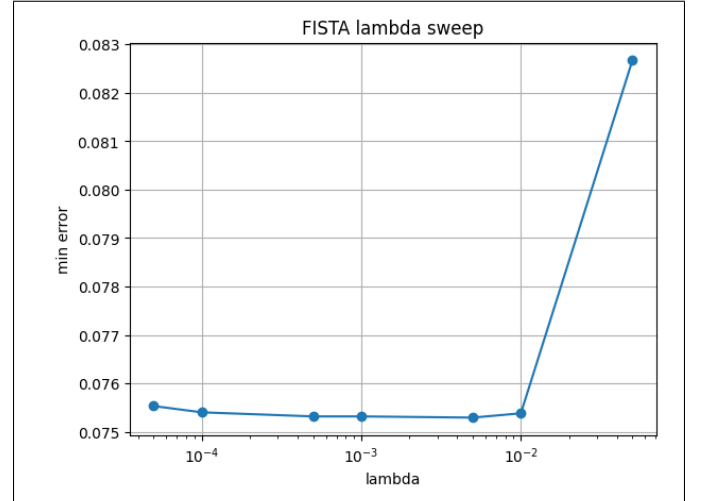


Fig. 4. Relative error of the FISTA model for different Lambda values

Performance of the computational imaging system was evaluated by comparing the output for the POCS model, a standard FISTA implementation, and our proposed FISTA + PnP model. These models were assessed using two different metrics: the speed of convergence (measured in iterations), and the reconstruction accuracy (based on the relative error calculation). The result from this experiment are summarized in Table 1 below.

	Convergence (iterations)	Relative Error
POCS	20	0.096
FISTA	300	0.075
FISTA + PnP	30	0.067

5.1 Convergence Analysis

The standard POCS model converges in the least amount of iterations (≈ 20), and serves as the fastest computation model for the problem. FISTA + PnP does not take much longer to converge at ≈ 30 iterations (Note, while it takes less iterations, FISTA + PnP takes much more time per iteration due to needing to use a contractive denoiser). Finally, the FISTA model takes close to 300 iterations before it converges.

5.2 Image Quality

The FISTA + PnP model surpasses both benchmarks in improving accuracy (30.2% and 10.6% reduction in error for the POCS and FISTA respectively). Notably, some of the finer detail in the images are much easier to make out in the reconstructed image using the FISTA + PnP model.

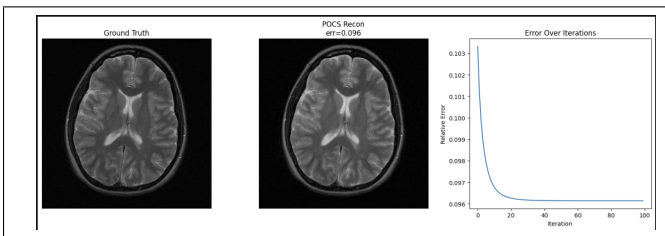


Fig. 5. Results from the reconstruction of an MRI Image using the POCS algorithm

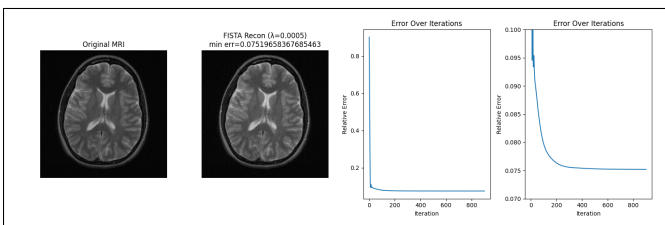


Fig. 6. Results from the reconstruction of an MRI Image using the FISTA algorithm

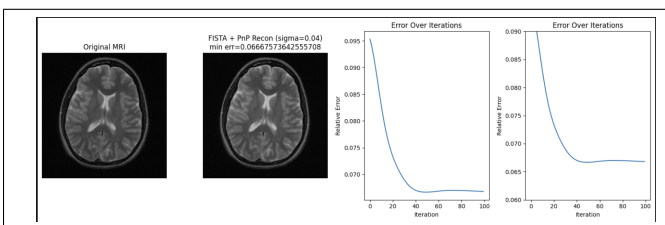


Fig. 7. Results from the reconstruction of an MRI Image using the FISTA + PnP algorithm

Overall, the data suggests that the FISTA + PnP helped improve the overall reconstruction of MRI images, but suffers a little due to requiring more computation time.

6 CONCLUSION

In this work, we investigated the optimization of compressive MRI reconstruction by altering different aspects of the model. We incorporated different denoising and inverse solving strategies to optimize image reconstruction. While it was not the fastest model, the FISTA + PnP model, with an integrated contractive denoiser, was able to achieved both a good convergence rate and superior relative error, leading to a more accurate reconstruction of the image. This model serves as a strong foundation for use as a High-Resolution MRI imaging tool. It is best used in scenerios where structural precision is more important than immediate feedback. This model is still able to leverage the under-sampling of the k-space to speed up scanning times, while leveraging a slightly longer computation time in order to create a close reconstruction of the original image. Primarily, this aids in detecting subtle changes while still ensuring a fast scan time. For instance, for Pediatric MRI's, kid struggle to stay still for a long period of time. Normal MRI would take a very long time to run, while the POCS model, while scanning much faster, will have some error tied to it. Our FISTA + PnP model can still leverage the shorter scan time from under-sampling while getting a clearer result. Moving forward, a natural extension of our work could involve finding a way to optimize the denoising stage in the model. A significant portion of the slower convergence is due to the fact the current PnP denoiser (trained on a large selection of image) is not Lipschitz continuous. Furthermore, the denoising model used (DruNet) was trained on a large variation of imaging data not limited to just MRI. Training a stronger denoising model on MRI images, and putting a stronger enforcement to ensure the PnP model stays Lipschitz continuous could help improve the MRI reconstruciton. Alternatively, the PnP model could be used alongside a different inverse solving algorithm in order to improve accuracy and convergence. For instance, implementing uncontrolled networks alongside FISTA would allow for the model to learn more from the image on each iteration. Creating a model with dynamic parameters would serve to improve training, which could aid in improving the overall quality of a compressive MRI image.

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- [3] [Deepinv Pretrained Models](#)
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